

Activities Firefox Web Browser May 22 14:18

about:sessionrestore x Home x 23521830_NguyenHoangVy Browsing HDF5 x + -

localhost:8888/notebooks/BTTH_Spark%2FDS200%2FLab-04%2F23521830_NguyenHoangVy_Lab04 80% ☆

jupyter 23521830_NguyenHoangVy_Lab04 Last Checkpoint: 3 minutes ago

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JupyterLab Python 3 (ipykernel)

```
ui_customers = spark.read.csv(BASE_PATH + "Customers.csv", header=True, inferSchema=True, sep=";")
df_orders = spark.read.csv(BASE_PATH + "Orders.csv", header=True, inferSchema=True, sep=";")
df_products = spark.read.csv(BASE_PATH + "Products.csv", header=True, inferSchema=True, sep=";")
df_order_items = spark.read.csv(BASE_PATH + "Order_Items.csv", header=True, inferSchema=True, sep=";")
df_order_reviews = spark.read.csv(BASE_PATH + "Order_Reviews.csv", header=True, inferSchema=True, sep=";")
```

2. Thống kê tổng số đơn hàng, số lượng khách hàng và người bán

```
[2]: total_orders = df_orders.count()
total_customers = df_customers.select("Customer_Trx_ID").distinct().count()
total_sellers = df_order_items.select("Seller_ID").distinct().count()

print("Tổng số đơn hàng:", total_orders)
print("Tổng số lượng khách hàng:", total_customers)
print("Tổng số lượng người bán:", total_sellers)
```

```
[Stage 19:>] (0 + 1) / 1
Tổng số đơn hàng: 99441
Tổng số lượng khách hàng: 99442
Tổng số lượng người bán: 3895
```

3. Phân tích số lượng đơn hàng theo quốc gia, sắp xếp theo thứ tự giảm dần

```
[3]: df_orders_cust = df_orders.join(df_customers, "Customer_Trx_ID", "inner")

df_orders_by_country = df_orders_cust.groupBy("Customer_Country") \
    .agg(count("Order_ID").alias("Tong_Don_Hang")) \
    .orderBy(col("Tong_Don_Hang").desc())

df_orders_by_country.show(truncate=False)
```

Activities Terminal May 22 14:13

vy23521830@UIT: ~

```
vy23521830@UIT:~$ hadoop-3.3.6/bin/hdfs dfs -put /media/sf_hadoop/DS200/Lab-04/Orders.csv /Lab04_DS
vy23521830@UIT:~$ hadoop-3.3.6/bin/hdfs dfs -put /media/sf_hadoop/DS200/Lab-04/Products.csv /Lab04_DS
vy23521830@UIT:~$ pyspark
[I 2026-05-22 14:12:41.230 ServerApp] jupyter_lsp | extension was successfully linked.
[I 2026-05-22 14:12:41.233 ServerApp] jupyter_server_terminals | extension was successfully linked.
[I 2026-05-22 14:12:41.237 ServerApp] jupyterlab | extension was successfully linked.
[I 2026-05-22 14:12:41.240 ServerApp] notebook | extension was successfully linked.
[I 2026-05-22 14:12:41.822 ServerApp] notebook_shim | extension was successfully linked.
[I 2026-05-22 14:12:41.887 ServerApp] notebook_shim | extension was successfully loaded.
[I 2026-05-22 14:12:41.889 ServerApp] jupyter_lsp | extension was successfully loaded.
[I 2026-05-22 14:12:41.890 ServerApp] jupyter_server_terminals | extension was successfully loaded.
[I 2026-05-22 14:12:41.985 LabApp] JupyterLab extension loaded from /home/vy23521830/.local/lib/python3.8/site-packages/jupyterlab
[I 2026-05-22 14:12:41.985 LabApp] JupyterLab application directory is /home/vy23521830/.local/share/jupyter/lab
[I 2026-05-22 14:12:41.986 LabApp] Extension Manager is 'pypi'.
[I 2026-05-22 14:12:41.983 ServerApp] jupyterlab | extension was successfully loaded.
[I 2026-05-22 14:12:41.986 ServerApp] notebook | extension was successfully loaded.
[I 2026-05-22 14:12:41.986 ServerApp] Serving notebooks from local directory: /home/vy23521830
[I 2026-05-22 14:12:41.986 ServerApp] Jupyter Server 2.14.2 is running at:
[I 2026-05-22 14:12:41.986 ServerApp] http://localhost:8888/tree?token=4d560ba27966082017efefed7dddcac350749501e717a20b
[I 2026-05-22 14:12:41.986 ServerApp] http://127.0.0.1:8888/tree?token=4d560ba27966082017efefed7dddcac350749501e717a20b
[I 2026-05-22 14:12:41.986 ServerApp] Use Control-C to stop this server and shut down all kernels (twice to skip confirmation).
[C 2026-05-22 14:12:42.093 ServerApp]

To access the server, open this file in a browser:
file:///home/vy23521830/.local/share/jupyter/runtime/jpserver-3850-open.html
Or copy and paste one of these URLs:
http://localhost:8888/tree?token=4d560ba27966082017efefed7dddcac350749501e717a20b
http://127.0.0.1:8888/tree?token=4d560ba27966082017efefed7dddcac350749501e717a20b
[I 2026-05-22 14:12:42.145 ServerApp] Skipped non-installed server(s): basedpyright, bash-language-server, dockerfile-language-server, r-nodesjs, javascript-typescript-langserver, jedi-language-server, julia-language-server, pyrefly, pyright, python-language-server, python-lsp-server, r-language-server, sql-language-server, texlab, typescript-language-server, unified-language-server, vscode-css-languageserver-bin, vscode-html-languageserver-bin, vscode-json-languageserver-bin, yaml-language-server
[W 2026-05-22 14:12:58.397 ServerApp] 404 GET /lab/api/settings/@jupyterlab/apputils-extension:kernel-settings?1779433978394 (127.0.0.1): Schema not found: /home/vy23521830/.local/share/jupyter/lab/schemas/@jupyterlab/apputils-extension/kernels-settings.json
[W 2026-05-22 14:12:58.398 LabApp] wrote error: 'Schema not found: /home/vy23521830/.local/share/jupyter/lab/schemas/@jupyterlab/apputils-extension/kernels-settings.json'
Traceback (most recent call last):
File "/home/vy23521830/.local/lib/python3.8/site-packages/tornado/web.py", line 1788, in _execute
    result = self._execute_request(request, **kwargs)
File "/home/vy23521830/.local/lib/python3.8/site-packages/tornado/web.py", line 1788, in _execute
    result = self._execute_request(request, **kwargs)
```

```
Activities Terminal May 22 14:19 vy23521830@UIT: ~  
python-lsp-server, r-languageserver, sql-language-server, texlab, typescript-language-server, unified-language-server, vscode-css-lan  
guageserver-bin, vscode-html-languageserver-bin, vscode-json-languageserver-bin, yamll-language-server  
[W 2026-05-22 14:12:58.397 ServerApp] 404 GET /lab/api/settings/@jupyterlab/apputils-extension:kernels-settings?1779433978394 (127.0  
.0.1): Schema not found: /home/vy23521830/.local/share/jupyter/lab/schemas/@jupyterlab/apputils-extension:kernels-settings.json  
[W 2026-05-22 14:12:58.398 LabApp] wrote error: 'Schema not found: /home/vy23521830/.local/share/jupyter/lab/schemas/@jupyterlab/app  
utils-extension:kernels-settings.json'  
Traceback (most recent call last):  
  File "/home/vy23521830/.local/lib/python3.8/site-packages/tornado/web.py", line 1788, in _execute  
    result = method(*self.path_args, **self.path_kwargs)  
  File "/home/vy23521830/.local/lib/python3.8/site-packages/tornado/web.py", line 3301, in wrapper  
    return method(self, *args, **kwargs)  
  File "/home/vy23521830/.local/lib/python3.8/site-packages/jupyterlab_server/settings_handler.py", line 58, in get  
    result, warnings = get_settings(  
  File "/home/vy23521830/.local/lib/python3.8/site-packages/jupyterlab_server/settings_utils.py", line 385, in get_settings  
    schema, version = _get_schema(schemas_dir, schema_name, overrides, labextensions_path)  
  File "/home/vy23521830/.local/lib/python3.8/site-packages/jupyterlab_server/settings_utils.py", line 61, in _get_schema  
    raise web.HTTPError(404, notfound_error % path)  
tornado.web.HTTPError: HTTP 404: Not Found (Schema not found: /home/vy23521830/.local/share/jupyter/lab/schemas/@jupyterlab/appu  
tils-extension:kernels-settings.json)  
[W 2026-05-22 14:12:58.405 LabApp] 404 GET /lab/api/settings/@jupyterlab/apputils-extension:kernels-settings?1779433978394 (759c349f  
ab764d58bee40a1602a1accb@127.0.0.1) 10.02ms referer=http://localhost:8888/tree  
[W 2026-05-22 14:13:22.920 ServerApp] Notebook BTTH_Spark/DS200/Lab-04/solution_lab4.ipynb is not trusted  
[W 2026-05-22 14:13:25.469 ServerApp] 404 GET /lab/api/settings/@jupyterlab/apputils-extension:kernels-settings?1779434005449 (127.0  
.0.1): Schema not found: /home/vy23521830/.local/share/jupyter/lab/schemas/@jupyterlab/apputils-extension:kernels-settings.json  
[W 2026-05-22 14:13:25.469 LabApp] wrote error: 'Schema not found: /home/vy23521830/.local/share/jupyter/lab/schemas/@jupyterlab/app  
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  File "/home/vy23521830/.local/lib/python3.8/site-packages/jupyterlab_server/settings_utils.py", line 385, in get_settings  
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  File "/home/vy23521830/.local/lib/python3.8/site-packages/jupyterlab_server/settings_utils.py", line 61, in _get_schema  
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tornado.web.HTTPError: HTTP 404: Not Found (Schema not found: /home/vy23521830/.local/share/jupyter/lab/schemas/@jupyterlab/appu  
tils-extension:kernels-settings.json)
```

```
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[W 2026-05-22 14:13:25.469 LabApp] wrote error: 'Schema not found: /home/vy23521830/.local/share/jupyter/lab/schemas/@jupyterlab/app  
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tils-extension:kernels-settings.json)  
[W 2026-05-22 14:13:25.472 LabApp] 404 GET /lab/api/settings/@jupyterlab/apputils-extension:kernels-settings?1779434005449 (759c349f  
ab764d58bee40a1602a1accb@127.0.0.1) 17.98ms referer=http://localhost:8888/notebooks/BTTH_Spark/DS200/Lab-04/solution_lab4.ipynb  
[W 2026-05-22 14:13:26.094 ServerApp] Notebook BTTH_Spark/DS200/Lab-04/solution_lab4.ipynb is not trusted  
[I 2026-05-22 14:13:32.570 ServerApp] Kernel started: bae0ab76-5f9a-4950-8fcc-52a71ca9deec  
26/05/22 14:13:40 WARN Utils: Your hostname, UIT resolves to a loopback address: 127.0.1.1; using 10.0.2.15 instead (on interface en  
p0s3)  
26/05/22 14:13:40 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address  
Setting default log level to "WARN".  
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).  
26/05/22 14:13:43 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where  
applicable  
[I 2026-05-22 14:13:46.213 ServerApp] Connecting to kernel bae0ab76-5f9a-4950-8fcc-52a71ca9deec.  
[I 2026-05-22 14:13:46.241 ServerApp] Connecting to kernel bae0ab76-5f9a-4950-8fcc-52a71ca9deec.  
[I 2026-05-22 14:13:46.276 ServerApp] Connecting to kernel bae0ab76-5f9a-4950-8fcc-52a71ca9deec.  
[I 2026-05-22 14:15:00.413 ServerApp] Saving file at /BTTH_Spark/DS200/Lab-04/23521830_NguyenHoangVy_Lab04.ipynb  
[I 2026-05-22 14:17:00.489 ServerApp] Saving file at /BTTH_Spark/DS200/Lab-04/23521830_NguyenHoangVy_Lab04.ipynb  
[I 2026-05-22 14:19:00.581 ServerApp] Saving file at /BTTH_Spark/DS200/Lab-04/23521830_NguyenHoangVy_Lab04.ipynb
```

```
Activities Terminal May 22 14:20 vy23521830@UIT: ~

return method(self, *args, **kwargs)
File "/home/vy23521830/.local/lib/python3.8/site-packages/jupyterlab_server/settings_handler.py", line 58, in get
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26/05/22 14:13:40 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
Setting default log level to "WARN".
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[I 2026-05-22 14:13:46.213 ServerApp] Connecting to kernel bae0ab76-5f9a-4950-8fcc-52a71ca9deec.
[I 2026-05-22 14:13:46.241 ServerApp] Connecting to kernel bae0ab76-5f9a-4950-8fcc-52a71ca9deec.
[I 2026-05-22 14:13:46.276 ServerApp] Connecting to kernel bae0ab76-5f9a-4950-8fcc-52a71ca9deec.
[I 2026-05-22 14:15:00.413 ServerApp] Saving file at /BTTH_Spark/DS200/Lab-04/23521830_NguyenHoangVy_Lab04.ipynb
[I 2026-05-22 14:17:00.489 ServerApp] Saving file at /BTTH_Spark/DS200/Lab-04/23521830_NguyenHoangVy_Lab04.ipynb
[I 2026-05-22 14:19:00.581 ServerApp] Saving file at /BTTH_Spark/DS200/Lab-04/23521830_NguyenHoangVy_Lab04.ipynb
^C[I 2026-05-22 14:19:53.370 ServerApp] interrupted
[I 2026-05-22 14:19:53.371 ServerApp] Serving notebooks from local directory: /home/vy23521830
1 active kernel
Jupyter Server 2.14.2 is running at:
http://localhost:8888/tree?token=4d560ba27966082017efefed7dddcac350749501e717a20b
http://127.0.0.1:8888/tree?token=4d560ba27966082017efefed7dddcac350749501e717a20b
Shut down this Jupyter server (y/[n])? y
[C 2026-05-22 14:19:56.128 ServerApp] Shutdown confirmed
[I 2026-05-22 14:19:56.147 ServerApp] Shutting down 5 extensions
[I 2026-05-22 14:19:56.148 ServerApp] Shutting down 1 kernel
[I 2026-05-22 14:19:56.168 ServerApp] Kernel shutdown: bae0ab76-5f9a-4950-8fcc-52a71ca9deec
vy23521830@UIT:~$
```